SHOVAN BERA

+91-8927967122 — dev.shovanbera@gmail.com — Barasat, Kolkata, West Bengal – 700125
linkedin.com/in/shovanbera — github.com/theshovan

PROFESSIONAL SUMMARY

B.Tech Computer Science (AI & ML) student with strong fundamentals in data structures, algorithms, object-oriented programming, and software design. Built full-stack and backend projects using Node.js/Express, Python/Flask, MongoDB, and vanilla JavaScript, plus NLP-based search and speech-processing applications. Comfortable with Git, RESTful APIs, SQL/NoSQL databases, and debugging, and eager to build AI-powered applications with agentic workflows and AI-assisted coding.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, Java
Backend & APIs: Node.js, Express.js, Flask, FastAPI, RESTful APIs, GraphQL APIs, JWT Authentication, Postman
Frontend: HTML5, CSS3, JavaScript
Databases: MongoDB (NoSQL), MySQL, SQL (Joins, Aggregations, Subqueries)
AI & ML: Machine Learning Fundamentals, NLP , Scikit-learn, Generative AI Fundamentals, AI APIs, AI-Assisted Coding
Data: Pandas, NumPy, Matplotlib, Jupyter Notebook, Data Cleaning, EDA
Tools & Practices: Git, GitHub, VS Code, SDLC, Debugging, OOP, Software Design Principles
Core CS: Data Structures & Algorithms, DBMS, Problem Solving

PROJECTS

Multilingual Video Transcription & Summarization — Python, Flask, SpeechRecognition, MoviePy, Pydub GitHub

- Built a Flask web app that extracts audio from uploaded video/audio files, transcribes long recordings in chunks, and returns transcripts and summaries in English, Bengali, and Hindi.
- Implemented language detection, translation, and an extractive summarization routine; exposed a REST endpoint returning JSON to an HTML front end.
- Debugged audio-processing, dependency, and deployment issues to improve reliability.

Smart ToDo API — Node.js, Express.js, MongoDB, JWT GitHub

- Developed a RESTful backend with secure signup/login using JSON Web Tokens and bcrypt password hashing.
- Implemented full CRUD task endpoints with authentication middleware protecting private routes; organized code into routes, controllers, models, and middleware layers with a MongoDB data model.

AetherStudy – Study Planner & Focus Dashboard — HTML5, CSS3, JavaScript GitHub

- Built a single-page app in modular vanilla JavaScript (planner, Pomodoro timer, flashcards, quiz hub) with shared state and a gamified XP/level/streak system.
- Added spaced-repetition active-recall pop-ups, Web Audio sound alerts, and HTML5 desktop notifications; persisted user data with browser Local Storage.

Spotify Lyric Search Engine — Python, Scikit-learn, Pandas, NLP GitHub

- Built an NLP search engine that identifies song title and artist from short lyric snippets across 57,000+ tracks using TF-IDF vectorization and cosine similarity.
- Created a preprocessing pipeline (tokenization, regex noise removal, stop-word filtering) and an evaluation script to test prediction accuracy.

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML), Brainware University, Kolkata Aug 2022 – Jul 2026
Higher Secondary Certificate (WBCHSE) 2022

CERTIFICATIONS & COMMUNITY

- Certifications: 5-Day AI Agents: Intensive Vibe Coding Course; Emerging Trends in Technology; Quantum Computing Workshop
- Active member of college tech clubs; regularly solve DSA problems in C++ and Python on LeetCode, HackerRank, and CodeChef; collaborate on team projects using Git.